

Birla Institute of Technology and Science Pilani (Rajasthan)
Department of Electrical and Electronics Engineering
EEE F313/ INSTR F313, Analog and Digital VLSI Design
First Semester

[Time: 90 min]

Date: 29 Oct 2021

Mid-semester Test (Open Book)

Max. Marks: 90

General Instructions to the candidate

- Please write your answers legibly and neatly in the answer booklet provided only.
- Symbols, constants, terminology used have their usual meaning unless specified specially.
- There are a total of **FIVE** questions and all are compulsory. Each carries marks as indicated.

Unless specifically mentioned—

- Use long channel approximation and lambda based design rules unless specially mentioned.
- Use square law current equation. Neglect body effect, channel length modulation if not mentioned specifically. Assume step input/s
- The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

Answer all the subparts of a question together in sequence.

1. (a) If the following Boolean expression is implemented using static CMOS logic, identify all the possible Euler's path. $Y = \overline{(AB + C)}(D + E)$

(b) A MOSFET is designed using 100 nm technology. It is now supposed to be redesigned using 22nm technology. The supply voltage for the 100nm device was 1.2 V. However, for the 22nm device, the supply voltage used is 0.6V. Answer the following questions assuming that devices are in short channel regime for both technologies.

- What happens to carrier velocity in the 22 nm scale device w.r.t 100 nm device? Indicate only constant/ increases/ decreases.
- If the drift current in the 100 nm device was 16 μA , the drift current in the 22nm device will be?
- Find the power density in the 22nm node given it was 12 mW/cm² in 100nm node.
- Find the ratio of gate delay in 22nm node with respect to the 100 nm node.
- Find the ratio of the energy delay product in 22nm node with respect to the 100 nm node.

[10 marks]

2. Consider the circuit in Fig. 1. The resistance per unit length of the interconnect lines is 1 k Ω /μm and capacitance is 0.5 pF/μm. The arrival time of inputs A, B and D is at $t = 1$ ns while the arrival time of input C is at 5 ns. Delay of the AND gates in 180nm technology node is 10ns and delay of buffer is 1ns.

(a) Identify the critical path through the given circuit and calculate the worst-case delay from input to output Y.

(b) Compute the optimum number of repeaters to be used in the critical path to arrive at the lowest possible delay. Here, keep the delay of the gates unchanged. Now compute the minimum delay through the critical path after introduction of repeaters.

(c) Justify if the delay of the critical path can be further reduced by changing the placement of inputs of the circuit obtained in part (b). If yes, what is the value of the modified delay after changing input placement.

[15 marks]

3. A CMOS inverter is designed in a process to achieve the lowest average gate delay. (Here t_{pLH} need not be equal to t_{pHL}). Given for this technology, $R_{eqn} = 15k\Omega$, $R_{eqp} = 45k\Omega$, $W_{min} = 450nm$, $L_{min} = 250nm$, $C_{ox} = 10 pF/\mu m^2$, $C_{load} = 15.15pF$ and best possible stage effort $p = 4$.

(a) Compute the optimum (W/L) ratio of pMOS to nMOS for this technology.

(b) Calculate the input capacitance C_{in} of a minimum sized inverter.

$$C_{in} = 3.7 \times \sqrt{\frac{R_{eqn}}{R_{eqp}}} = \beta_{opt}$$

(c) Compute the logical effort and parasitic of the gates mentioned in the table. Redraw the table and populate it with the calculated values.

Table-1		
Gate	g	p
NOT Gate		
2 input NOR gate		
2input NAND gate		
3 input NOR gate		

(d) Consider the circuit given in Fig. 2. Using the values calculated in part (c), compute the optimum delay from input to output along the path of interest shown (Here, keep the no. of stages fixed)

(e) For Fig. 2, compute the number of stages for minimum delay. Hence modify and redraw the circuit to achieve this delay.

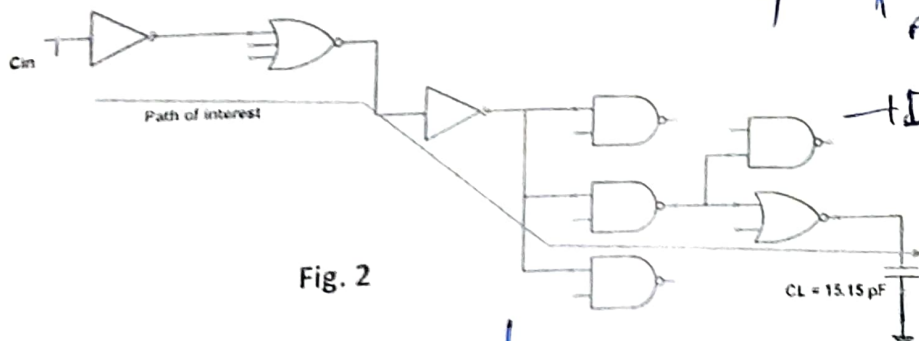


Fig. 2

[25 marks]

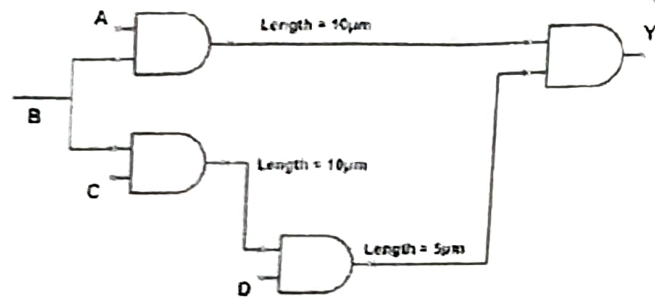


Fig. 1

An inverter is shown in the Fig.3.

Use the following parameters $\mu_n C_{ox} = 0.6 \text{ mA/V}^2$, $\mu_p C_{ox} = 0.2 \text{ mA/V}^2$, $V_{T0N} = |V_{T0P}| = 1 \text{ V}$, C_{out} (including all parasitic capacitances) = 200fF.

- (a) Find the aspect ratio of the PMOS such that the switching threshold is exactly 2.5V.

For sub-parts (b) to (e), use $(W/L)_{PMOS} = (W/L)_{NMOS} = 10$

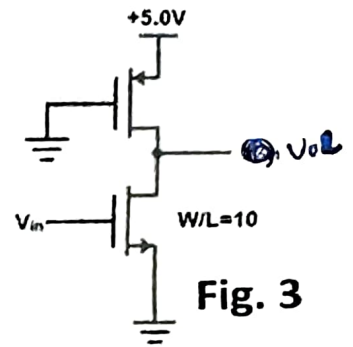
- (b) Calculate the Noise Margin Low (NM_L).

- (c) Calculate the maximum value of static power dissipation in the inverter.

- (d) Calculate the worst-case propagation delay using *average current* method.

- (e) What is the dynamic power dissipation of a 7-stage ring oscillator built by repeatedly cascading this inverter? Assume the ring oscillator is generating a perfect square wave.

[30 marks]



5. An embedded microprocessor core is integrated in a system-on-chip, which is designed in 180 nm process, 1.8V supply and has 10 million logic transistors and 20 million memory transistors with an average width of 12λ and 4λ respectively. The gate capacitance, $C_g = 2 \text{ fF}/\mu\text{m}$. The logic gates have an activity factor of 0.5 whereas memory transistors have an activity factor of 0.25.

- (a) Calculate the maximum clock frequency if the dynamic power should not exceed 20 mW?

- (b) In this 180nm process, the subthreshold leakage is 20 nA/ μm and the gate leakage is 2 nA/ μm . If half of the transistors are off on average, estimate the static power consumption.

- (c) Assume that the input signal is having a rise/ fall time of 2ps out of which 0.5ps results in peak current drawn from supply is 0.2mA. Assume operating frequency of 200MHz with activity factor of 0.5 for logic block with only 10^3 gates and 0.25 for memory block with only 10^2 gates, find out the total short circuit power dissipation.

[10 marks]

END